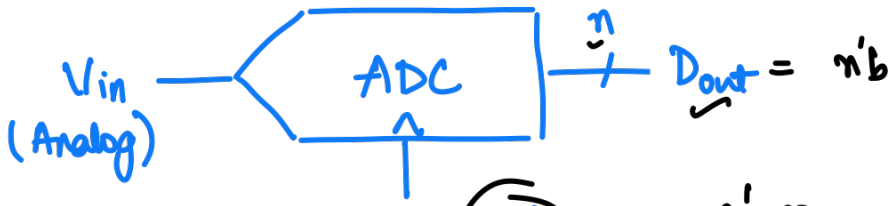
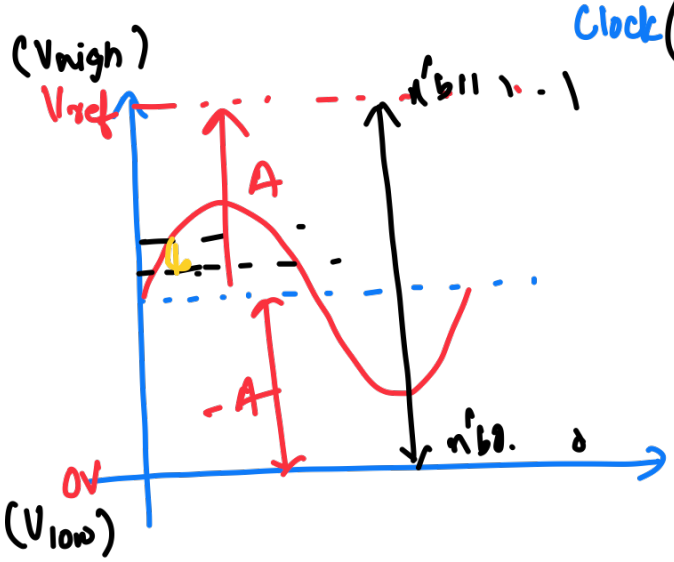
Lecture # 38

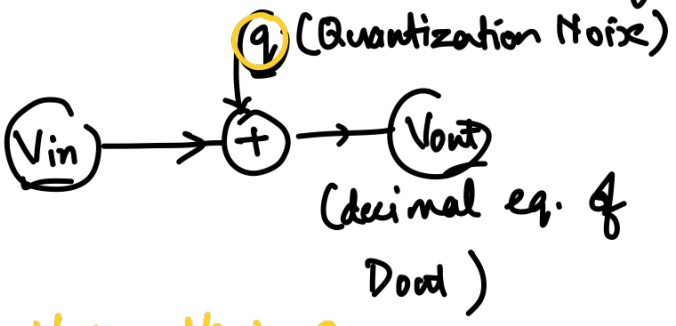
- SNR (Signal-to-Noise Ratio)



$n'b00\ldots000 = V_{low}$

$n'b11\ldots111 = V_{high}$





$V_{out} = V_{in} + q$

$$V_{out} = \left(2^0 b_0 + 2^1 b_1 + \ldots + 2^{n-1} b_{n-1}\right) \frac{V_{ref}}{2^n}$$

$$-\frac{\Delta}{2} < q < \frac{\Delta}{2}, \quad \Delta = \frac{V_{ref}}{2^n}$$

- $E[q^2] = \frac{\Delta^2}{12}$
- $E[q] = 0$
- $V_{in} = A \sin(\omega_{in} t)$

$$\overline{V_{in}^2} = \frac{A^2}{2}$$

$$\text{Signal-to-Noise Power} = \frac{\text{Signal Power}}{\text{Noise Power}}$$

$$= \frac{\text{Signal Power}}{\text{Quantization Noise Power}}$$

$$= \frac{A^2/2}{\Delta^2/12}$$

$$\Delta = \frac{2A}{2^n}$$

$$= \frac{A^2/2}{\frac{4A^2}{2^{2n}} \times \frac{1}{12}}$$

0.4771
0.3010

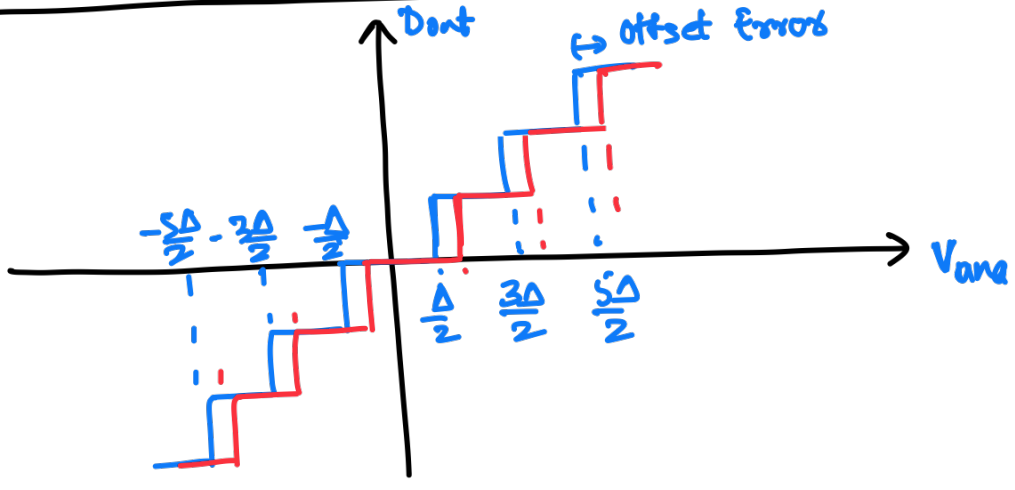
$$SNR = 3 \times 2^{2n-1}$$

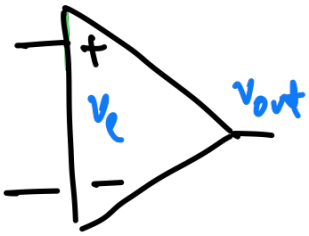
$$SNR \text{ [in dB]} = 10 \log_{10}\left(3 \times 2^{2n-1}\right)$$

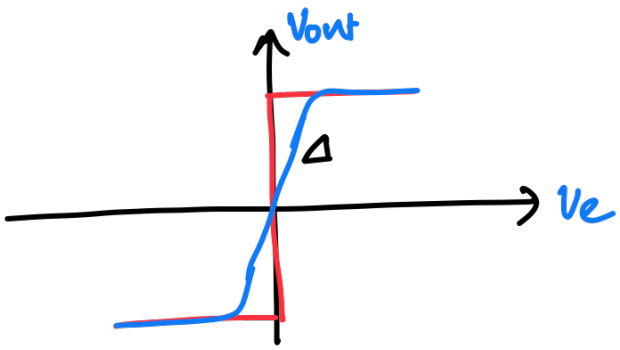
$$= 10 \log_{10}\left(1.5 \times 2^{2n}\right)$$

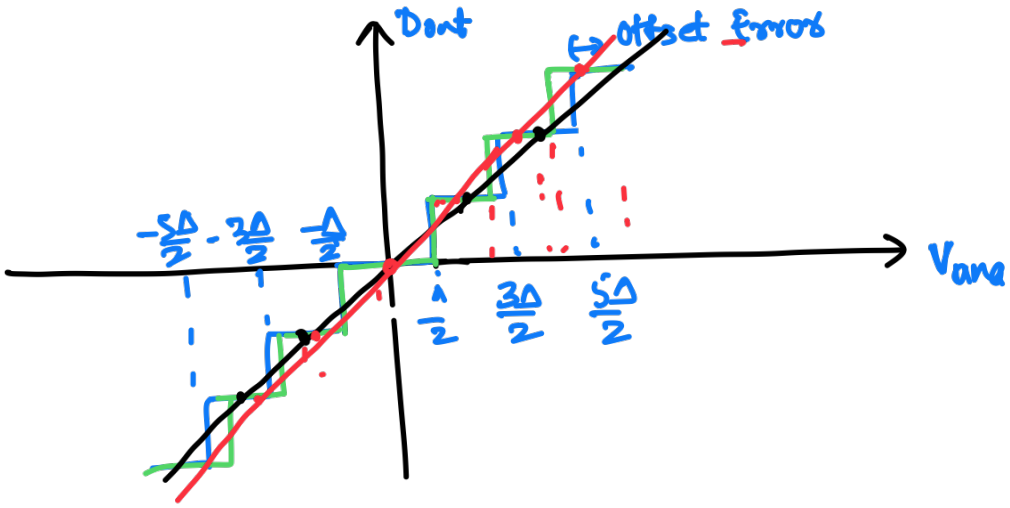
$$SNR = 6.02n + 1.76 \quad \text{[in dB]}$$

- 10-bit ADC
- 61.96 dB .
- Sampling speed, f_s
  - 100 MS/s - 10-bit ADC
  - 1MS/s , 23-bit ADC.
  - 20 MHz , 4 bit ADC









$D_{out} = V_{ana} + Q.$

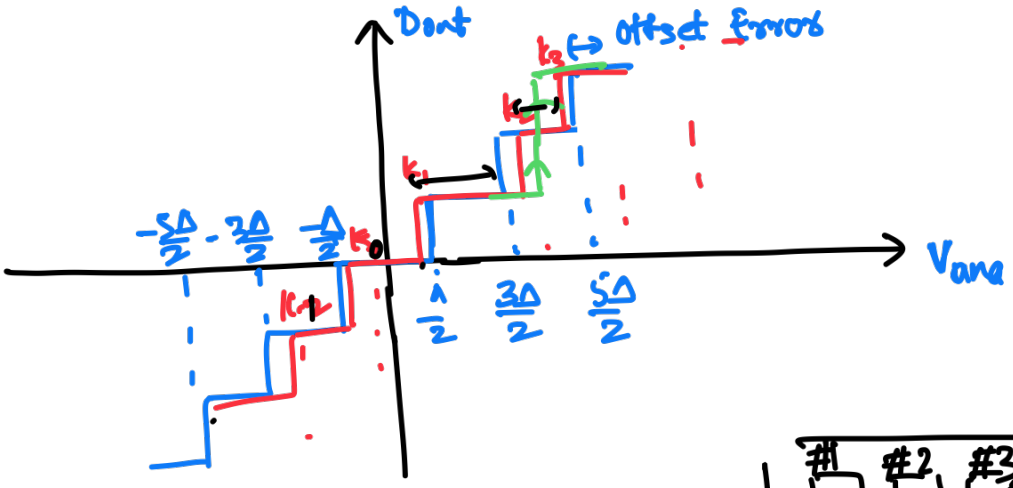
(decimal eq.)

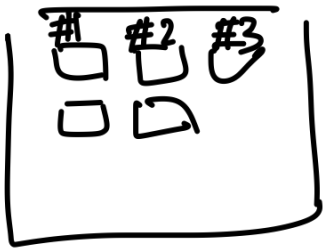
$D_{out.} = K V_{ana} + Q$

(Gain Errors)

$K \neq 1$

- Gain error and offset errors can be estimated and compensated.





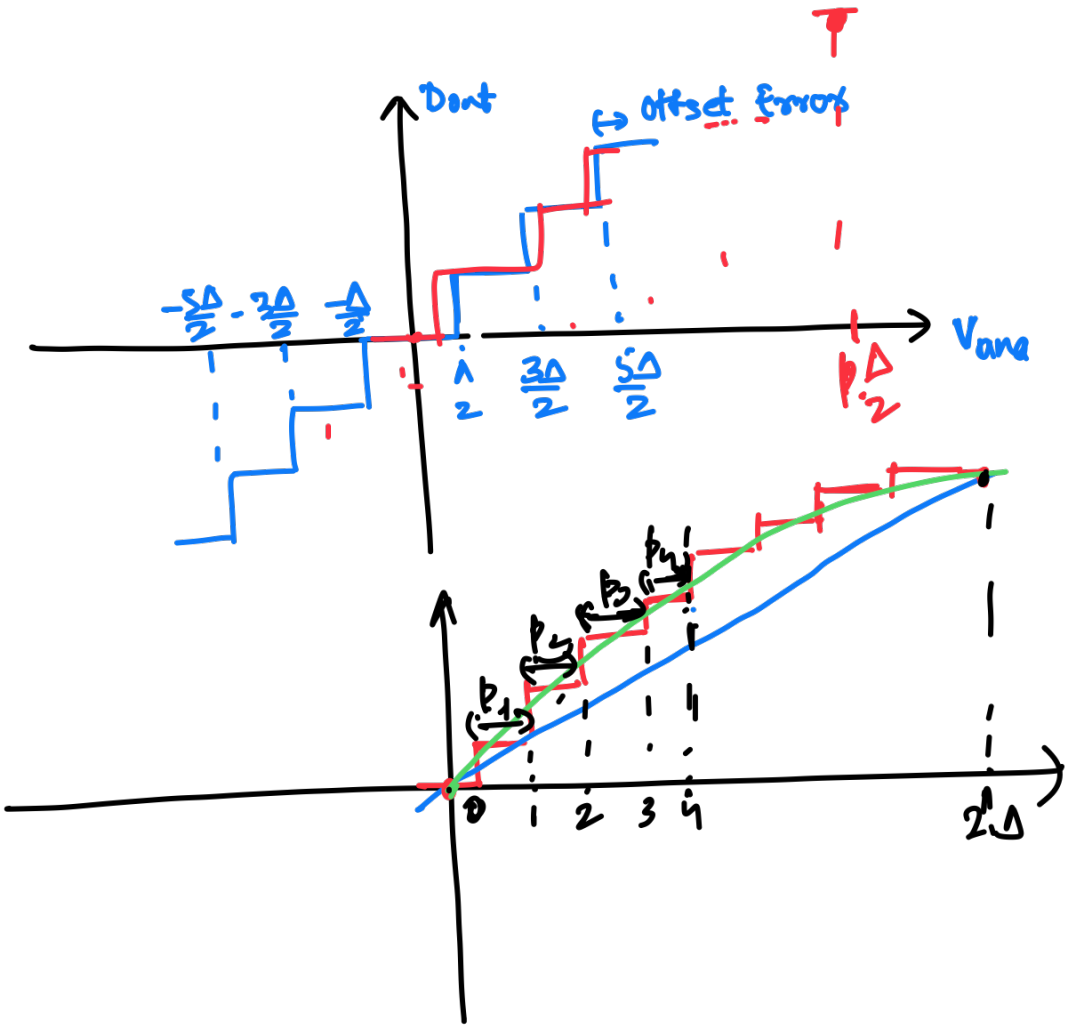
- Differential Non-linearity (DNL)

$$DNL(k_1) = \frac{(V_{ana,k_1} - V_{ana,k_0}) - \Delta}{\Delta} =$$

$$= \frac{(V_{ana,k_1} - V_{ana,k_0})}{\Delta} - 1\,LSB$$

$$= \frac{\frac{V_{ana,k_1} - V_{ana,k_0}}{\Delta} - 1\,LSB}{1\,LSB}$$

$$|DNL| < 0.5\,LSB$$



Integral Non-linearity (INL) = $\sum$

$$INL(i) = \sum_{i=0}^{i-1} p_i - i$$